

Element H Report

Test Criteria & Procedure

Problem Statement

Reports indicate that 42,000 American homeowners annually face recurring neck, shoulder, and back injuries from picking up and disposing of leaves in their yards¹. Disregarding this task causes leaves to accumulate in yards, smothering the grass and blocking the sunlight needed for proper growth. Overall, leaf removal is a task that can cause physical strain when done and, if neglected, can affect the yard's health, leading to grass death in the fall.

Design Specifications

The specifications listed below are set for the lawn sweeper. Changes have been made to the ranking of certain requirements as the design has changed after brainstorming a design. Every specification will not be tested. Each specification is listed in importance, 1 being the most important and 16 being the least important.

1. Efficiency
 - a. The machine shall pick up leaves from a 2,500 sq. ft. lawn within 60 minutes under weather conditions such as clear sky or some wind.
2. Leaf Disposal
 - a. The machine shall autonomously pick up leaves and dispose of them in its own storage unit.
3. Battery Life
 - a. The machine's battery shall last 120 minutes and recharge in 50 minutes
4. Battery Replacement
 - a. The machine shall feature a swappable battery that can be replaced in under five minutes.
5. Storage
 - a. The machine shall have sufficient storage to accommodate at least 30 gallons of leaves with a weight capacity of 50 lbs.
6. Compaction
 - a. The machine shall compact collected leaves by a ratio of 1.5:1 in volume
7. Battery Per Runtime

¹ Don't let raking leaves become a pain in the neck, back or spine. Luminis Health. (n.d.). <https://www.luminishealth.org/en/blog/dont-let-raking-leaves-become-a-pain-in-the-neck-back-or-spine?>

- a. The machine shall operate over a distance of 7,500 square feet in one charge, as according to conducted research, that is the average lawn size for suburban homes in the U.S.
- 8. Target Cost
 - a. The machine shall not exceed the sale cost of \$550
- 9. Noise Level
 - a. The machine shall not exceed 60 dB during operation, measured at a distance of 5 feet.
- 10. Durability
 - a. The machine shall be operational and efficient for at least eight fall seasons.
- 11. Ergonomics
 - a. Any human interference shall minimize awkward lifting positions and not exceed 20 pounds.
- 12. Autonomous Operation
 - a. The machine shall operate fully autonomously once it is set up and trained
- 13. Operating Environment
 - a. The machine shall operate effectively outdoors in temperatures ranging from 30 degrees Fahrenheit to 75 degrees Fahrenheit.
- 14. Natural Conditions
 - a. The machine shall operate with resistance to conditions like dust, moisture, and uneven terrain
- 15. Safety Precautions
 - a. The machine shall have an emergency stop in the event of malfunctions, debris clogging, or unexpected terrain.
- 16. Maintenance
 - a. The machine's maintenance shall be limited to battery replacement and minimal inspection once per fall season.

Criteria	Description of data required	Quantitative or qualitative	Unit of measurement
Specification #1: Efficiency	Volume of leaves picked up in a certain amount of time	Quantitative	Gallons/ Hour
Specification #5: Storage	Gallons of leaves is the max capacity of the machine.	Quantitative	Gallons
Specification #11: Ergonomics	User feedback on comfort when using the product	Qualitative	----
Specification #14: Natural Conditions	How well the machine was able to pick up leaves in conditions with wet leaves, sticks, etc.	Qualitative	----

Specifications #2-4 7,12, 15 & 16 will not be tested because the design is manual, not requiring a battery for operation.

Specification #6 will not be tested because the design does not feature a compacting component in the storage.

Specification #8 will not be tested because the cost of materials was predetermined from the bill of materials.

Specification #9 will not be tested because it was determined that the noise level is not something that impacts the user experience drastically.

Specification #10 will not be tested because of the imitations in the testing timeline.

Specification #13 will not be tested because those conditions cannot be accurately replicated during testing.

Specification #1: Efficiency

Names:

- Shreyas Mathur

Team Members:

- Seth Jones
- Tanish Lakshmanan

Incremental Testing Summary:

- No incremental testing is required for this test. The test evaluates the Lawn Sweeper's efficiency after the prototype construction is complete.

Testing Location:

- Tanish's Backyard
- Neuqua Valley High School

Purpose

- The purpose of this test is to determine the efficiency of the Lawn Sweeper by measuring the volume of leaves it can collect within a 30-minute time frame. The results of this test will be collected in gallons/hour to determine its efficiency.

Pass/Fail Criteria

- Pass: This test is successful if...
 - The Lawn Sweeper consistently collects more than 15 gallons of leaves in the allotted 30-minute time frame
 - The machine is undamaged and ready for repeated use after a trial is complete
- Fail: This test is unsuccessful if...
 - The Lawn Sweeper collects an insufficient volume of leaves within the allotted time
 - The Lawn Sweeper breaks down between/within trials and cannot perform properly

Materials

- Lawn Sweeper Prototype
- Leaves
- Measuring Container for Leaves
- Timer/Stopwatch
- Data Sheet

Initial Conditions

- The Lawn Sweeper prototype is fully built and operational
- The testing area has a known volume of leaves
- Leaves are evenly distributed across the testing area

Safety Concerns

Hazards	Controls
Jam fingers in the axle	Clear the test area of large debris and uneven surfaces
Running over the foot with the product	Close-toe footwear
	Not to get close to moving parts of the machine

Test Termination

- Terminate this procedure if any of the following conditions occur:
 - The prototype gets damaged while being used
 - Outdoor Conditions are not optimal for testing

Data:

<u>Test #</u>	<u>Volume of Leaves Collected</u>	<u>Time Elapsed</u>	<u>Efficiency (Gallons/Hour)</u>

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Distribute leaves evenly across the testing area
5. Start the timer and operate the machine for a set period of time.
6. Measure the total volume of leaves at the end of the test
7. Calculate Efficiency in gallons/hour.
8. Record all data on the accompanying datasheet.

Approval Signature: Shreyas Mathur

Specification #5: Storage

Names:

- Tanish Lakshmanan

Team Members:

- Seth Jones
- Shreyas Mathur

Incremental Testing Summary:

- N/A

Testing Location:

- Tanish's Backyard
- Neuqua Valley High School

Purpose

- The purpose of this test is to determine the storage capacity of the device by measuring the maximum volume of leaves the storage container can hold before it can no longer hold any more leaves. The results will be recorded in gallons.

Pass/Fail Criteria

- Pass: This test is successful if...
 - Leaves are securely contained within the storage compartment
 - The storage compartment remains structurally intact and usable after testing.
- Fail: This test is unsuccessful if...
 - Leaves spill during use.
 - The storage compartment breaks or detaches.

Materials

- Lawn Sweeper Prototype
- Leaves
- Data Sheet
- Device to measure leaf volume
- Stopwatch/ Timer

Initial Conditions

- The Lawn Sweeper prototype is fully built
- The storage compartment is empty at the start

Safety Concerns

Hazards	Controls
Jam fingers in the axle	Not to get close to moving parts of the machine
Running over the foot with the product	Close-toe footwear

Test Termination

- Terminate this procedure if any of the following conditions occur:
 - The prototype gets damaged while being used
 - The weather conditions are interfering with results

Data:

<u>Test #</u>	<u>Volume of Leaves Collected</u>	<u>Time Till Storage Full</u>

Stepwise Procedure

1. Ensure all materials are available
2. Verify all initial conditions are true
3. Review all safety precautions
4. Confirm the storage container is empty
5. Start the stopwatch and push the lawn sweeper across the lawn to collect leaves
6. Stop the stopwatch and stop collecting when the lawn sweeper reaches its maximum capacity
7. Measure the total volume of leaves collected
8. Record data on the data sheet

Approval Signature: Tanish Lakshmanan

Specification #11: Ergonomics

Names:

- Seth Jones

Team Members:

- Shreyas Mathur
- Tanish Lakshmanan

Incremental Testing Summary:

- No incremental testing is required for this test. The test evaluates the Lawn Sweeper's user comfort after the prototype construction is complete.

Testing Location:

- Neuqua Valley High School

Purpose

- The purpose of this test is to determine if the Lawn Sweeper is comfortable and ergonomic for the user to operate and find any flaws that may cause discomfort or difficulty. Results will be from a range of participants who will use the design and give their input on comfort, ease of use, and overall experience.

Pass/Fail Criteria

- Pass: This test is successful if..
 - Most of the participants who tested the product say it was designed well, comfortable to use, and provided other positive feedback from a survey.
- Fail: The test is unsuccessful if:
 - Most of the participants find the product uncomfortable to use, and have feedback on how the design should be changed and improved.

Materials

- Participants
- A survey
- Lawn Sweeper Prototype
- Leaves
- Lawn Area
- Documentation
- Data sheet

Initial Conditions

- The Lawn Sweeper prototype is fully built and operational
- 20 participants are willing to participate in the test

Safety Concerns

Hazards	Controls
Jam fingers in the axle	Instruction of how to safely use the product
Run over foot with product	Closed-toe footwear

Test Termination

- Terminate this procedure if any of the following conditions occur:
 - The prototype gets damaged while being used
 - Participant feels any major discomfort when using the product

Data:

Qualitative data will be collected by a survey of participants who use the lawn sweeper. The survey will be a free response, asking the participants how their experience was and any feedback they have for the product. Questions that will be asked are:

- How comfortable was the product in use? 1 being mostly uncomfortable, 5 being mostly comfortable.
- Please describe how comfortable/uncomfortable using the product was.
- Any feedback or major concerns?

<u>Test #</u>	<u>Comfort of Product (1-5)</u>	<u>Feedback on the product</u>

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with how to use the prototype.
4. Test the product, letting participants use the prototype in the designated lawn area
5. Have all participants fill out the survey
6. Repeat the test with at least 20 participants
7. Record all data and survey responses on a spreadsheet

Approval Signature: Seth Jones

Specification #14: Natural Conditions

Names:

- Shreyas Mathur

Team Members:

- Seth Jones
- Tanish Lakshmanan

Incremental Testing Summary:

- No incremental testing is required for this test. The test evaluates the Lawn Sweeper's performance under real outdoor conditions after the prototype construction is complete.

Testing Location:

- Tanish's Backyard
- Neuqua Valley High School

Purpose

- The purpose of this test is to evaluate the Lawn Sweeper's efficiency under real-world conditions that the machine may face while working. This includes assessing how effectively the lawn sweeper collects leaves when they are dry and wet, as well as determining how common outdoor obstacles, such as small branches or sticks, affect its operation. The results of this test will help determine the Lawn Sweeper's flexibility in less-than-ideal situations.

Pass/Fail Criteria

- Pass: This test is successful if:
 - The Lawn Sweeper can pick up above 70% of the leaves under at least 2 tested conditions on the first try.
 - The machine operates properly when small obstacles are present
- Fail: This test is unsuccessful if:
 - The Lawn Sweeper fails to pick up more than 70% of leaves under the tested conditions
 - Obstacles like small sticks prevent the machine from operating properly
 - The machine gets clogged with leaves and requires intervention while operating.

Materials

- Lawn Sweeper Prototype
- Wet & Dry leaves
- Lawn Area
- Small Sticks or branches
- Data Sheet
- Camera for documentation

Initial Conditions

- The Lawn Sweeper prototype is fully built and operational
- Weather conditions are appropriate for outdoor testing

Safety Concerns

Hazards	Controls
User tripping over branches or uneven surfaces	Close-toe footwear
Pinching fingers in the machine	Not to get close to moving parts of the machine

Test Termination

- Terminate this procedure if any of the following conditions occur:
 - The machine gets damaged
 - Weather conditions become unsafe
 - There is a safety risk

Data:

<u>Test #</u>	<u>Testing Condition</u> <u>(Dry/Wet/</u> <u>Sticks)</u>	<u>Total Leaves</u>	<u>Leaves</u> <u>picked up on</u> <u>the first try</u>	<u>% of leaves</u> <u>picked up</u>	<u>Notes:</u>

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. Test the lawn sweeper on dry leaves and record observations.
4. Test the lawn sweeper on wet leaves and record observations.
5. Place small sticks or branches among the leaves and test again.
6. Observe how well the sweeper collects leaves in each condition.
7. Repeat the test in each condition 5 times.
8. Record all qualitative and quantitative data on the datasheet.

Approval Signature: Shreyas Mathur